Project Charter: Kaudi (Cowrie Shell Game)

Centurion University of Technology and Management (CUTM)
Center for Indigenous Knowledge & Heritage Game Development
Collection: Native & Indigenous Sports of India — Title 18

Charter Field	Project Specification
Project Name	**Kaudi: The Sacred Cowrie Shell Scoring Game** (and Shared Collection Dice Engine)
Project Description	Kaudi is a fast-paced, tactile indigenous Indian scoring game played with natural cowrie shells (*kauri*). Shells are tossed onto a heritage woven mat, and points are scored depending on the orientation of the mouths (smooth dorsal vs. serrated aperture). The digital implementation features 2–4 player local pass-and-play, tactical AI opponents, a "Press-Your-Luck" risk-reward mode, authentic hand-crafted Indian visual aesthetics, and serves as the master, deterministic Cowrie Shell RNG core powering the entire Indigenous Indian Board Game Collection (including *Ashtapada*, *Chauka Bara*, *Pachisi*, and *Pasha Palli*).
Project Sponsor	**Centurion University of Technology and Management (CUTM)**
Department of Sports & Cultural Heritage / Game Development Studio	
Anthology Reference: "An Anthology of Native & Indigenous Sports of India" (CUTM, 2025)	
Project Team Member(s)	• **Project Lead & Technical Director**: Swarup & Core Team
• **Game Designer & Cultural Lead**: Abhi Mitra (Lead Designer)	
• **Lead Software Engineer**: Rules Engine & WebGL/HTML5 Core Developer	

Charter Field	Project Specification
• **Art & UI/UX Specialist**: Heritage Asset & Typography Lead	
• **Audio & Sound Designer**: Acoustic Heritage SFX Lead	
• **QA & Playtest Coordinator**: Automated Rules & Test Matrix Lead	
Project Objectives	1. **Cultural Preservation & Authenticity**: Recreate the authentic tactile sensation, scoring rules, and historical ambiance of traditional Indian cowrie gambling/scoring games as documented in historical texts and CUTM field research.

2. **Deterministic Shared Engine**: Author an engine-agnostic, mathematically sound, seedable Cowrie Dice Rules service (`CowrieDice`) that outputs true binomial shell distributions, grace throws, and special bonuses for reuse across all Tier-2 race games.

3. **High Accessibility & Responsive Design**: Deliver a lightweight (<15MB initial download, <3s load time) WebGL and HTML5/Canvas responsive application supporting touch, keyboard, and mouse across mobile, tablet, and desktop screens.

4. **Engaging Gameplay & Solvers**: Implement both Classic Target Race (first to 100) and high-stakes Press-Your-Luck modes accompanied by an intelligent Expected-Value (EV) driven AI heuristic with adjustable difficulty levels (Easy, Medium, Hard). |
| **Project Milestones** | • **M0 — Cowrie Core Engine (Week 1.0)**: Pure rules module, seeded RNG, 4/6/7 shell probability matrix, specials/grace detection, unit test coverage (1M roll verification).
• **M1 — Visuals & Toss Animation (Week 1.5)**: 2D physics-inspired throw dynamics, scatter & settle motion, 128px shell assets, mouth-up/mouth-down sprite switches.
• **M2 — Game Loops & AI Logic (Week 2.0)**: Turn state machine, scoring trays, press-your-luck bank/bust loop, heuristic AI agent with 3 risk profiles.
• **M3 — UI/UX & Design Bible Compliance (Week 2.5)**: Mukta/Tiro Devanagari typography, Centurion University heritage color palette (teak, sand, ink), 88px touch targets, audio synthesis.
• **M4 — WebGL & PWA Release (Week 3.0)**: Responsive canvas, offline Service Worker, PWA manifest, itch.io & CUTM portal embedding, zero-soft-lock QA sign-off. |
| **Assumptions** | 1. Players of all ages can immediately grasp cowrie count mechanics with zero prior tutorial required.

5. Target hardware includes budget Android smartphones, web browsers (Chrome, Safari, Firefox, Edge), and classroom interactive displays.

6. The shared dice engine architecture will directly integrate into companion collection titles without refactoring core dice logic.

7. Cultural accuracy takes precedence over modern fantasy gamification tropes; assets must strictly reflect Indian folklore aesthetics. |

| **Project Completion Date** | **Target Production Release**: October 2026
Current Milestone Active: Alpha Engine & Interactive Web Verification |

Project Authorization & Sign-Off

We agree that this is a viable project. The Project Team Members are authorized to begin the planning, design, and development process for this project in accordance with Centurion University guidelines and the Native & Indigenous Sports of India master vision.

Role	Name	Signature / Status	Date
Project Sponsor	CUTM Dean of Heritage & Technology	*Approved*	2026-09-29
Lead Game Designer	Abhi Mitra	*Approved*	2026-09-29
Lead Developer	Swarup / Lead Programmer	*Active*	2026-09-29
QA Lead	Automated Test Matrix Coordinator	*Approved*	2026-09-29